

Development of 5-Axis Linear Motor Driven Super-Precision Machine

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Abstract

Recent strong demand for production of ultra-precision components drives the need for 5-axis super-precision machines. A design study of such a 5-axis super-precision machine, which is capable of nanometer-level precision cutting, has been conducted by searching innovative solutions to the three major problems: friction, thermal deformation, and positioning accuracy. The final design implemented the following: stiff aerostatic guideways for near frictionless motion while structural rigidity was maintained, thermally stable structures made of alumina ceramics, and smooth control of linear motor based actuation system by employing high resolution and highly responsive linear encoders and linear power amplifiers. The designed machine has been built to verify the targeted performance. Various shapes that require nanometer-accuracy have been created using the machine. The results have consistently indicated stable nanometer-level accuracy. Targeted design performance has been successfully demonstrated.

Keywords:

Ultra-precision, Nano technology, Linear drive

1 INTRODUCTION

The recent rapidly growing needs in the mass production of very high precision components, which are referred to as opto-mechatronics components for consumer products, require the new type of ultra precision machine for the machining of the molds for such very high precision components. Figure 1 shows an example of such components used in opto-mechatronics systems, namely, the binary optics lens used in a CD/DVD system. This type of lens requires 10-nanometer to nanometer accuracy and precision over the entire three-dimensional complex geometry. (More details of such optical lens can be found in [1].) Machining of the mold for components with such a sophisticated shape requires a versatile 5-axis super-precision machining center. Considering that the final machined components should satisfy 10-nanometer to nanometer level accuracy and precision, and the fact that some uncertainty and disturbances exist in the machining process, such a 5-axis machining center is required to have subnanometer to a nanometer motion accuracy and precision.

Although there have been several past research works on the development of high precision machine tools, most of them have been developed for the machining of simple flat or circular geometry with 1 to 3 axes combined motion achieving typically several ten nanometers level motion accuracy and precision [2-6]. There have also been past researches on the high precision 5-axis machine, but the machine developed could achieve the motion accuracy and precision up to only several tens of nanometers [2, 7].

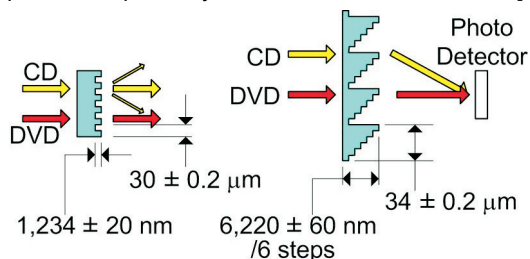


Figure 1: Geometry for binary optics lens for CD/DVD.

The purpose of this study is to develop a 5-axis super-precision machining system by examining the possible design approach and available technologies.

2 DESIGN APPROACH

In designing the system, possible countermeasures to overcome three major problems (friction, thermal deformation, and control and drive stability) have been carefully studied.

The design of the machine structure has been determined by two key characteristics: the minimum thermal deformation and static and dynamic stability when the machine is in motion. One of the unique targets of the machine performance is to achieve the high speed motion such as 3 m/min. with the controlled motion accuracy and precision of within one nanometer. Additionally, in order to reduce the energy cost of operation of the system, the machine should achieve such motion precision and machining speed even while the controlled temperature fluctuates within a 0.1 °C range. Movable components should also be as light as possible for better dynamic response characteristics.

By considering the requirements mentioned above as well as the manufacturability and properties of structure material, it has been concluded that the entire structure should be made of alumina ceramics.

For the machine guideways, in order to eliminate the nonlinear friction problem of contact type guideways, it has been decided that a non-contact type, aerostatic bearing should be implemented. Since the guideway is also made of alumina ceramics, the guideway system becomes very durable against the adherence of the bearing to the guide surface, which may possibly be caused by a touch down accident due to the sudden stop of air supply.

As for the feed drive and control system, a coreless linear motor drive system has been adopted because the drive system also needs to be non-contact in conjunction with the non-contact guideways. Coreless linear motors can eliminate the cogging problem, which has been one of the major unwanted nonlinear characteristics affecting the stability of drive systems. With coreless linear motors, smooth motion with better accuracy is possible.

The type of amplifiers has carefully been examined and the decision reached was to adopt linear amplifiers instead of PWM amplifiers. In nanometer-level motion control, a slight sudden fluctuation of power supply to the linear motor acts as noise to the thrust force. The latter section of this paper will show the difference in machine performance between the two types of amplifiers.

High resolution feedback sensors must be used as the linear motor based motion control system can never be more accurate than the resolution of the feedback sensor. In the study, two types of feedback sensor were examined: 1-nm resolution linear scale and 0.07-nm resolution laser linear scales. The laser scale system adopted is based on the grating interferometer together with volume holography in the diffraction grating. The scale system is accompanied by the specially designed interface circuit to allow high speed processing to achieve 3 m/min with 0.07 nanometers resolution. Again, the latter section of this paper will show the remarkable difference in the motion control performance of the two. The 0.07 nanometer resolution scale gives the machine the highest positioning resolution compared to other developments in past researches [4-7].

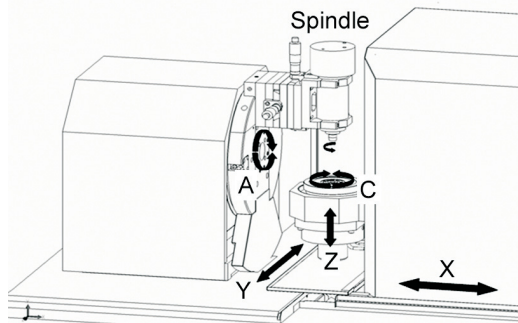


Figure 2: Final axis configuration of the machine.

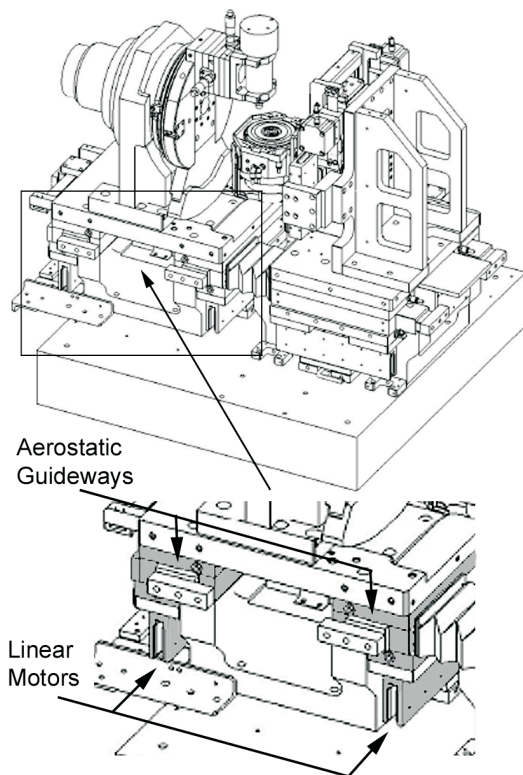


Figure 3: Machine structure, with details on the Y-axis.

3 THE DEVELOPED SUPER-PRECISION MACHINE

The machine configuration was designed for minimum error as well as for versatile functionality. To avoid the accumulation of error, the numbers of the axes that are stacked together were minimized as shown in Figure 2. The required function of the machine also determined where the axes were to be located. The workspace of the machine was designed to be 100x100x55 mm³. The Y-axis was designed to be independent of the X-axis such that motion error in one axis does not affect the other axis. The rest of the axes were arranged in such a way that the machine can perform basic operations (milling and turning) as well as complex machining of free form surfaces.

More details on the entire machine configuration are shown in Figure 3. Note that although only the Y-axis is shown in details, all other axes have a similar arrangement. To minimize pitch, roll and yaw motion in each axis, a pair of linear motors is placed along the guideways such that the combined two thrust force vector of the pair of linear motors always acts at the center of the moving mass.

4 BASIC MACHINE PERFORMANCE EXPERIMENTS

The primary objectives of the experiments are to study and confirm the performance of the designed machine.

4.1 Motion control performance

In order to study the variation of the motion control performance with respect to the different types of actuator power supply and the sensor resolutions, the comparative experiments have been performed.

The following three cases were examined:

Case 1: PWM power amplifier (switching frequency: 13.5 kHz) and 1-nm resolution scale; Case 2: linear power amplifier and 1-nm resolution scale; and Case 3: linear amplifier and 0.07-nm resolution scale.

In the experiments, the machine was given a series of 1 nm step position command along the Y-axis in both forward and backward direction motions.

The comparison between the experimental results of the two amplifier types (Figures 4 and 5) showed that the use of linear amplifiers dramatically reduced the positioning errors to approximately one-third of that when the PWM amplifiers were used.

As for the scale's resolution, the 0.07-nm resolution scale also effectively reduced the positioning errors to only approximately one-half to one-third of that when the 1-nm resolution scale was used. (Figures 5 and 6)

From these experimental results, it has been confirmed that the developed machine could achieve the targeted motion control accuracy of 1 nanometer when linear amplifiers and 0.07-nm resolution linear scales are used.

4.2 Temperature behavior of the machine and the environment

The machine was designed such that the targeted accuracy and precision can be maintained under the ambient temperature fluctuation range of 0.1°C. In order to check the thermal behavior of the machine, an experiment has been carried out by measuring the temperature at various locations in the machining system. Figure 7 shows the set-up locations of the temperature sensors and Figure 9 shows the results of the temperature measurements. Note that the temperature at the outlet of the temperature-controlled air chamber was also measured (not shown in Figure 7). In the measurement, after idling for two hours, the machine started to make repeated motions as shown in Figure 8

for approximately 3 hours, then returned to idling again for another 3 hours. Figure 9 shows such measurement results: the temperatures at all locations except at the linear motor coil stayed within 0.1°C range and only the temperature at the motor coil fluctuated within 0.2°C range when the controlled air temperature was maintained at $23.61 \pm 0.05^{\circ}\text{C}$. It has been concluded that the stable behavior within the targeted temperature fluctuation was achieved.

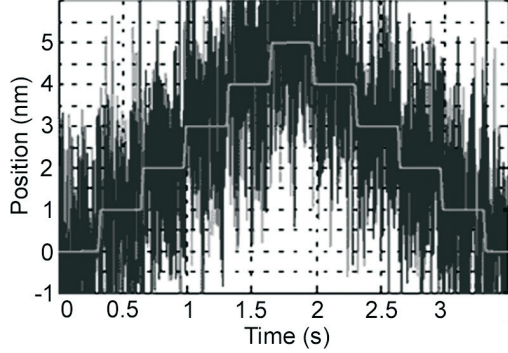


Figure 4: 1-nm step motion with PWM amplifiers and 1-nm resolution scales.

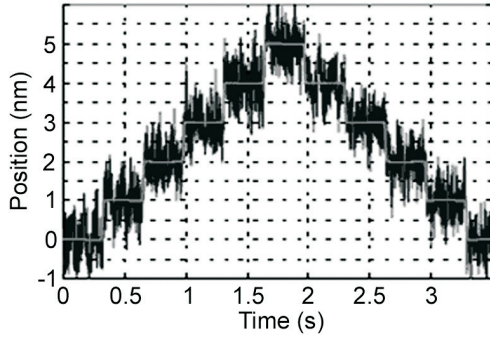


Figure 5: 1-nm step motion with linear amplifiers and 1-nm resolution scales.

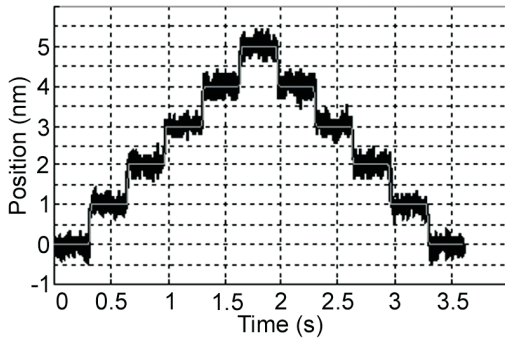


Figure 6: 1-nm step motion with linear amplifiers and 0.07-nm resolution scales.

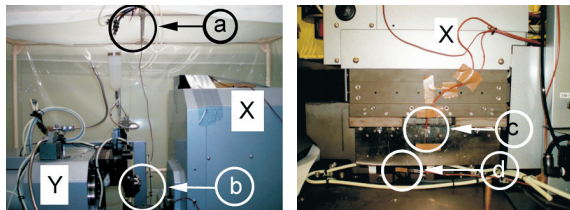


Figure 7: Temperature measurement positions. (a) At the top of the machine. (b) At the work jig. (c) At the X-axis motor's magnet. (d) At the X-axis motor's coil. (e) At the air outlet near the ceiling (not shown).

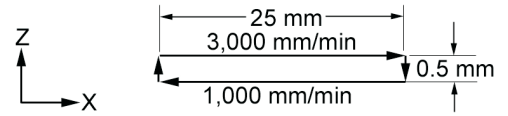


Figure 8: Machine motion during temperature measurement.

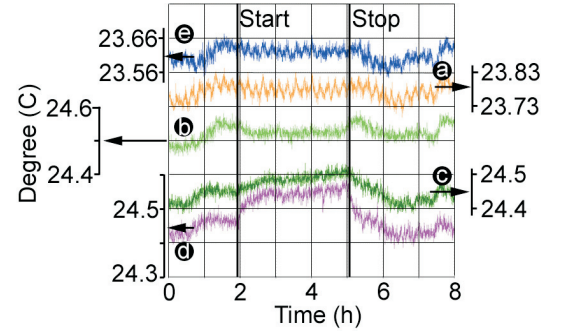


Figure 9: Temperature measured at various points.

4.3 Performance characteristics of the aerostatic guideways

The rigidity and damping are two major parameters governing the performance of the aerostatic bearing and these two parameters have an antithetical relationship. In a super-precision machining system, setting very high rigidity results in very low damping and the table tends to oscillate when a very small impulsive shock is given to the bearing system. By taking a heuristic approach, the rigidity was set optimally to the following values:

Vertical rigidity: $2,000 \text{ N}/\mu\text{m}$, Yawing moment rigidity: 110 Nm/sec , Pitching moment rigidity: 250 Nm/sec , Rolling moment rigidity: 530 Nm/sec with $5 \mu\text{m}$ bearing gap and 0.3 MPa supply air pressure.

5 MACHINING RESULTS

For confirming the overall machining performance, two different machining tests were conducted; nanostep and V-groove cuttings.

In a nanostep cutting test, steps of 10-nm depth were machined on an electroless plated Ni-P surface using a single crystal diamond shaping tool at the feedrates of $1,000$ and 300 mm/min for roughing and finishing, respectively. Figures 10 and 11 show the measurement results of machined surface by Zygo NV 5000. The micro region of the 10 nm step was also measured by using AFM (Atomic Force Microscope) as shown in Figures 12 and 13 in order to obtain more detailed profile of the machined surface. These results verify that the machine can provide the targeted machining accuracy and precision.

In a V-groove cutting test, a 300-nm deep, asymmetrical, V-groove was cut on a Ni-P plated workpiece surface using a single crystal diamond shaping tool at the feedrates of 300 and 50 mm/min for roughing and finishing, respectively. Such an asymmetrical V-groove with 10 nanometer form accuracy is one of the most critical geometry in producing opto-mechatronics components. The biggest challenge is in the fact that the V-groove has to be made by shaping each side of the groove separately by tilting the tool, as shown in Figure 14. With this method, if the machine accuracy and precision is not sufficient, the two sides of the groove cannot come to meet exactly where the sharp angle is intended. This means that the V-groove shape cannot be formed.

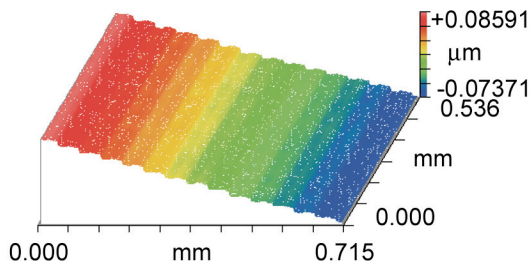


Figure 10: 10-nm step cutting results.

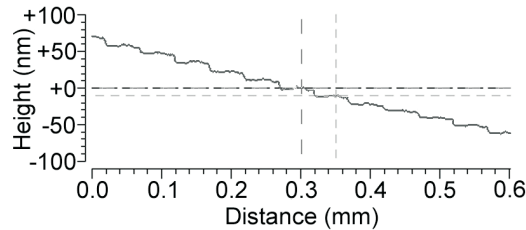


Figure 11: 10-nm step profile.

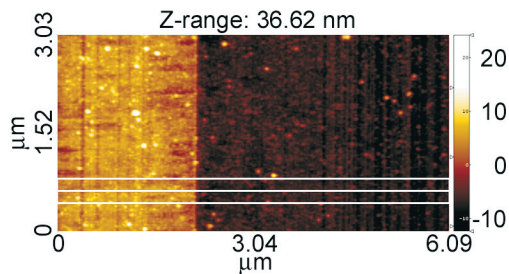


Figure 12: 10-nm step surface scanned by the AFM.

The results of the V-groove cutting (Figures 15 and 16) indicated excellent machining performance.

6 SUMMARY

- A 5-axis super-precision machine was developed to answer the growing need in applications that require complex free-form machining with 10-nanometer to nanometer precision and accuracy, e.g. molds for lens used in opto-mechatronics devices.
- During the development, the critical technologies to achieve the targeted performance of the system were identified. These technologies were:
 - The use of alumina ceramics for the construction of the machine for thermal stability under the controlled ambient temperature fluctuation of 0.1 °C, which is achievable as confirmed by experiment.
 - The use of stiff aerostatic guideways to provide noncontact motion.
 - The use of coreless linear motors together with linear amplifiers to provide smooth and virtually noise-free motion. When used together with high resolution (0.07 nm) linear scales, dramatic improvement in motion control performance was observed.
- It has been confirmed that the motion accuracy of 1 nm of the developed machine could consistently be achieved when the machine's ambient temperature was maintained within 1°C range. The results of the nanostep and V-groove cutting showed excellent machining performance.

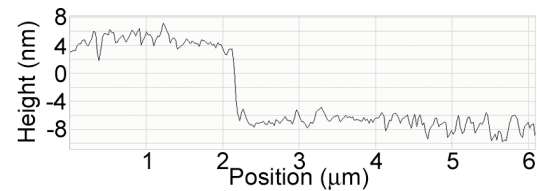


Figure 13: 10-nm step profile, as measured by the AFM.

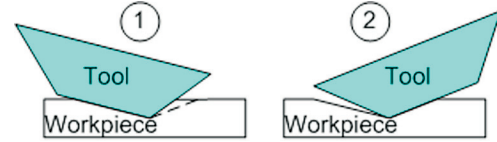


Figure 14: V-groove cutting process.

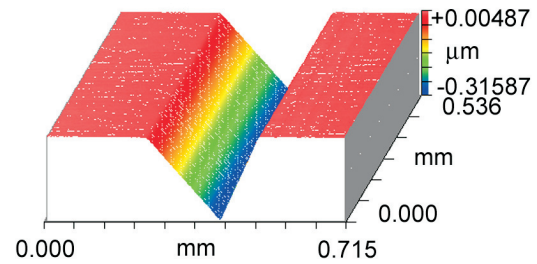


Figure 15: V-groove cutting result.

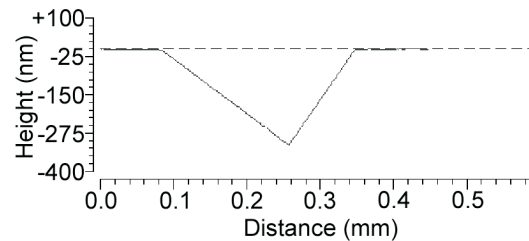


Figure 16: V-groove profile.

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